Precalculus with Limits, 2e: Duolingo Math checklist

Tick each textbook section as you read it and each Duolingo unit as you finish it. Units are listed per lane (a Duolingo grade or topic) in the order the app presents them, so you can follow one lane top to bottom. Ratings: full = covers the section's main skill, partial = covers part of it, prereq = a warm-up from an earlier grade. The right column names the textbook sections the unit serves.

Chapter 1: Functions and Their Graphs (p. 1)

Grade 9 carries the basics (functions, lines, transformations), grade 11 repeats them at precalculus depth, and the Algebraic Graphing topic is a condensed walk through the whole chapter. Follow grade 11 if you only have time for one lane.

Textbook sections

☐	1.1 Rectangular Coordinates	p. 2
☐	1.2 Graphs of Equations	p. 13
☐	1.3 Linear Equations in Two Variables	p. 24
☐	1.4 Functions	p. 39
☐	1.5 Analyzing Graphs of Functions	p. 54
☐	1.6 A Library of Parent Functions	p. 66
☐	1.7 Transformations of Functions	p. 73
☐	1.8 Combinations of Functions: Composite Functions	p. 83
☐	1.9 Inverse Functions	p. 92
☐	1.10 Mathematical Modeling and Variation	p. 102

Book only, not on Duolingo: **1.1** Translating points in the plane; Applications of the distance and midpoint formulas. **1.2** Symmetry tests (x-axis, y-axis, origin); Sketching graphs of general equations by point plotting. **1.3** Slope as a rate of change in applications. **1.4** The difference quotient; Writing a function from a word problem. **1.5** Even and odd functions; The Vertical Line Test. **1.6** The square root and reciprocal parent functions in detail. **1.8** Domain of a composite function; Decomposing a function into two simpler functions. **1.9** One-to-one functions and the Horizontal Line Test. **1.10** Inverse, joint, and combined variation; Least squares regression on the calculator.

Grade 6

☐	33. Plot points in all quadrants	prereq	1.1

Grade 7

☐	7. Proportionality in equations	prereq	1.10

Grade 8

☐	21. Intro to slope	prereq	1.3
☐	22. Slope formula	prereq	1.3
☐	24. Slope-intercept form	prereq	1.3
☐	28. Equations of parallel lines	full	1.3
☐	29. Equations of perpendicular lines	full	1.3
☐	36. Intro to functions	prereq	1.4

Grade 9

☐	5. Slope-intercept form of linear equations	full	1.3
☐	7. Standard form of linear equations	full	1.3
☐	17. Inputs and outputs	full	1.4
☐	18. Graphs of functions	full	1.5
☐	20. Vertical intercept of functions	partial	1.2
☐	21. Horizontal intercept of functions	partial	1.2
☐	23. Average rate of change	full	1.5
☐	24. Linear functions	prereq	1.6
☐	25. Domain of a function	full	1.4
☐	26. Range of a function	partial	1.4
☐	27. Graphs of piecewise functions	full	1.5
☐	28. Equations of piecewise functions	full	1.4
☐	30. Step functions	full	1.6
☐	31. Absolute value functions	full	1.6
☐	33. Graphs of inverse functions	partial	1.9
☐	34. Equations of inverse functions	partial	1.9
☐	35. Graphs of inverse linear functions	prereq	1.9
☐	36. Equations of inverse linear functions	prereq	1.9
☐	75. Vertical translations	full	1.7
☐	76. Horizontal translations	full	1.7
☐	77. Vertical stretches	full	1.7
☐	78. Vertical compressions	full	1.7
☐	79. Horizontal stretches	full	1.7
☐	80. Horizontal compressions	full	1.7
☐	81. Scatter plots	partial	1.10
☐	82. Correlation	partial	1.10
☐	83. Linear models	partial	1.10
☐	84. Residuals	partial	1.10

Grade 10

☐	10. Reflectional symmetry	prereq	1.2
☐	45. Lines in point-slope form	full	1.3
☐	46. Lines in different forms	full	1.3
☐	47. Circles on the plane	full	1.2
☐	50. Line segment midpoint	full	1.1
☐	51. Distance formula	full	1.1

Grade 11

☐	2. Function families	full	1.6
☐	3. Compare function graphs	full	1.5
☐	9. Transform absolute value functions	full	1.7
☐	13. Relative maxima and minima of polynomial functions	partial	1.5
☐	65. Inverse function equations	full	1.9
☐	66. Inverse function graphs	full	1.9
☐	67. Combine functions	full	1.8
☐	68. Function composition	full	1.8
☐	69. Vertical translations	full	1.7
☐	70. Horizontal translations	full	1.7
☐	71. X-axis function reflections	full	1.7
☐	72. Y-axis function reflections	full	1.7
☐	73. Vertical stretches	full	1.7
☐	74. Vertical compressions	full	1.7
☐	75. Horizontal stretches	full	1.7
☐	76. Horizontal compressions	full	1.7

Grade 12

☐	13. Function composition	full	1.8
☐	14. Inverse functions	full	1.9
☐	17. Domain and range	full	1.5
☐	18. Graphing inverses	full	1.9

Geometry

☐	11. Absolute value and distance	prereq	1.1

Algebraic Graphing

☐	3. Slope	partial	1.3
☐	4. Slope-intercept lines	partial	1.3
☐	8. Function notation	full	1.4
☐	9. Piecewise-defined functions	full	1.4
☐	10. Step and absolute value functions	full	1.6
☐	23. Inverse linear functions	partial	1.9
☐	24. Function translations	full	1.7
☐	25. Vertical function scaling	full	1.7
☐	26. Horizontal function scaling	full	1.7
☐	27. Reflections of functions	full	1.7

Statistics

☐	15. Linear models	partial	1.10
☐	16. Linear model fitness	partial	1.10

Chapter 2: Polynomial and Rational Functions (p. 125)

Grades 11 and 12 between them cover polynomial, complex, and rational functions thoroughly. Duolingo has no unit on polynomial long or synthetic division (2.3) or on nonlinear inequalities (2.7); use the book for those.

Textbook sections

☐	2.1 Quadratic Functions and Models	p. 126
☐	2.2 Polynomial Functions of Higher Degree	p. 136
☐	2.3 Polynomial and Synthetic Division	p. 150
☐	2.4 Complex Numbers	p. 159
☐	2.5 Zeros of Polynomial Functions	p. 166
☐	2.6 Rational Functions	p. 181
☐	2.7 Nonlinear Inequalities	p. 194

Book only, not on Duolingo: **2.1** Completing the square to write vertex form; Maximum and minimum word problems. **2.2** Using the Intermediate Value Theorem to locate a zero. **2.3** Polynomial long division; Synthetic division; The Remainder Theorem and the Factor Theorem. **2.4** Plotting complex numbers in the complex plane. **2.5** The Fundamental Theorem of Algebra; Descartes's Rule of Signs; Upper and lower bounds for real zeros. **2.6** Applications of rational functions. **2.7** Polynomial inequalities; Rational inequalities; The test-interval (sign chart) method.

Grade 9

☐	11. Solve one-variable inequaltiies	prereq	2.7
☐	12. Compound 'or' inequalities	prereq	2.7
☐	13. Compound 'and' inequalities	prereq	2.7
☐	56. Tables of quadratic functions	full	2.1
☐	57. Graphs of quadratic functions	full	2.1
☐	58. Equations of quadratic functions	full	2.1
☐	59. Zeros of quadratic functions	full	2.1
☐	60. Factored form	full	2.1
☐	61. Vertex	full	2.1
☐	62. Vertex form	full	2.1

Grade 11

	12. Intro to polynomial functions	full	2.2
	13. Relative maxima and minima of polynomial functions	full	2.2
	14. Zeros of polynomial functions	full	2.2, 2.5
	15. Multiplicity of polynomial functions	full	2.2
	16. End behavior of odd polynomial functions	full	2.2
	17. End behavior of even polynomial functions	full	2.2
	18. The imaginary number, i	full	2.4
	19. Square roots of negative numbers	full	2.4
	20. Patterns of exponentiating i	full	2.4
	21. Multiply complex numbers	full	2.4
	22. Complex roots	full	2.5
	23. Complex conjugates	full	2.4
	24. Intro to rational functions	full	2.6
	25. Y-intercept of rational functions	full	2.6
	26. X-intercepts of rational functions	full	2.6
	27. Vertical asymptotes of rational functions	full	2.6
	28. Rewrite rational functions	full	2.6
	29. End behavior approaching a horizontal line	full	2.6
	30. End behavior approaching y=0	full	2.6
	31. End behavior approaching slant asymptotes	full	2.6

Grade 12

	2. Polynomial function graphs	full	2.2
	3. Roots of polynomial functions	full	2.5
	4. Rational root theorem	full	2.5
	5. Polynomial function equations from graphs	full	2.2
	6. Vieta's formula for quadratic polynomials	partial	2.5
	7. Vieta's formula for cubic polynomials	partial	2.5
	8. Zeros of rational functions	full	2.6
	9. Vertical asymptotes of rational functions	full	2.6
	10. Horizontal asymptotes of rational functions	full	2.6
	11. Holes in rational functions	full	2.6
	12. Slant asymptotes of rational functions	full	2.6
	48. Intermediate value theorem	partial	2.2

Algebraic Graphing

☐	11. Intro to quadratic functions	full	2.1
☐	12. Quadratic graphs and equations	full	2.1
☐	13. Factored form of quadratics	full	2.1
☐	14. Vertex form of quadratics	full	2.1
☐	30. Polynomial function equations	full	2.2
☐	31. Polynomial function graphs	full	2.2
☐	32. Polynomial graphs and roots	full	2.2
☐	33. Rational function equations and graphs	full	2.6
☐	34. Rewrite and graph rational functions	full	2.6
☐	35. Rational function asymptotes	full	2.6
☐	36. Holes and slant asymptotes	full	2.6

Chapter 3: Exponential and Logarithmic Functions (p. 215)

Grade 11 covers exponentials and logarithms almost section by section. Duolingo stops at growth and decay, so the Gaussian, logistic, and logarithmic models of 3.5 come from the book only.

Textbook sections

☐	3.1 Exponential Functions and Their Graphs	p. 216
☐	3.2 Logarithmic Functions and Their Graphs	p. 227
☐	3.3 Properties of Logarithms	p. 237
☐	3.4 Exponential and Logarithmic Equations	p. 244
☐	3.5 Exponential and Logarithmic Models	p. 255

Book only, not on Duolingo: **3.1** Compound interest, n times per year and continuously. **3.4** Logarithmic equations; Checking for extraneous solutions. **3.5** Gaussian models; Logistic growth models; Logarithmic models (pH, Richter scale, decibels).

Grade 9

☐	70. Exponential function tables	prereq	3.1
☐	71. Exponential function graphs	prereq	3.1
☐	72. Exponential function equations	prereq	3.1

Grade 11

☐	38. Exponential function tables	full	3.1
☐	39. Exponential function graphs	full	3.1
☐	40. Exponential function equations	full	3.1
☐	41. Exponential growth	full	3.1, 3.5
☐	42. Exponential decay	full	3.1, 3.5
☐	48. Intro to the natural base	full	3.1
☐	49. Continuous growth models	full	3.1, 3.5
☐	50. Intro to logarithms	full	3.2
☐	51. Logarithm value	full	3.2
☐	52. Product rule	full	3.3
☐	53. Quotient rule	full	3.3
☐	54. Power rule	full	3.3
☐	55. Change of base formula	full	3.3
☐	56. Intro to logarithmic functions	full	3.2
☐	57. Logarithmic functions in base 10	full	3.2
☐	58. Logarithmic functions in base 2	full	3.2
☐	59. Logarithmic function graphs	full	3.2
☐	60. Logarithmic function intercepts	full	3.2
☐	61. The natural logarithm	full	3.2
☐	62. Natural logarithm graphs	full	3.2
☐	63. Solve exponential equations using logarithms	full	3.4
☐	64. Solve base e equations using logarithms	full	3.4

Grade 12

☐	15. Properties of logarithms	full	3.3
☐	16. Undo exponentials	partial	3.2, 3.4

Algebra

☐	18. Logarithms	full	3.2
☐	19. Product and quotient rules	full	3.3
☐	20. Properties of logarithms	full	3.3
☐	21. Solve logarithmic and exponential equations	full	3.4

Algebraic Graphing

☐	15. Intro to exponential functions	full	3.1
☐	16. Exponential graphs and equations	full	3.1
☐	18. Exponential growth and decay	partial	3.5
☐	19. The natural base, e	full	3.1
☐	20. Logarithmic functions	full	3.2
☐	21. Logarithmic function graphs	full	3.2
☐	22. Natural logarithmic function	full	3.2

Chapter 4: Trigonometry (p. 279)

If time is short, follow grade 10 for the right-triangle ratios and special-angle values of 4.3, then grade 11 into grade 12 for the sine, cosine, and tangent graphs of 4.5 and 4.6, where Duolingo is strongest. No unit covers radians, the unit circle, reference angles, cosecant, secant, cotangent, or inverse trig functions, so most of 4.1, 4.2, 4.4, and 4.7 comes from the book, with Duolingo giving only single steps and warm-ups. Section 4.8 gets the tools but not the applications: elevation and depression problems, bearings, and harmonic motion models are book only.

Textbook sections

☐	4.1 Radian and Degree Measure	p. 280
☐	4.2 Trigonometric Functions: The Unit Circle	p. 292
☐	4.3 Right Triangle Trigonometry	p. 299
☐	4.4 Trigonometric Functions of Any Angle	p. 310
☐	4.5 Graphs of Sine and Cosine Functions	p. 319
☐	4.6 Graphs of Other Trigonometric Functions	p. 330
☐	4.7 Inverse Trigonometric Functions	p. 341
☐	4.8 Applications and Models	p. 351

Book only, not on Duolingo: **4.1** Angles in standard position: vertex, initial side, terminal side; Positive (counterclockwise) and negative (clockwise) angles; Angles beyond one full revolution; Quadrant of an angle's terminal side; Coterminal angles in degrees and radians; Definition of radian measure (arc length over radius); Radian measures of common angles and quadrant boundaries; Acute and obtuse angles in radians (0 to pi/2, pi/2 to pi); Converting between degrees and radians; Degrees-minutes-seconds and decimal degrees; Complementary and supplementary angles in radians; Arc length formula s = r theta; Linear speed v = s/t, angular speed omega = theta/t, and v = r omega; Converting revolutions per minute to radians per minute; Area of a circular sector. **4.2** The unit circle and wrapping the real number line around it; Defining the six trigonometric functions from the point (x, y) for a real number t; Cosecant, secant, and cotangent values as reciprocals; Special-angle values in radians (pi/6, pi/4, pi/3), matched to the degree drills; Values at the quadrantal points 0, pi/2, pi, 3pi/2; Values and signs at standard points outside the first quadrant; Where tangent, cotangent, secant, and cosecant are undefined; Domain (all reals) and range [-1, 1] of sine and cosine; Using the 2pi period to evaluate at large or negative t; Even and odd trigonometric functions (cosine and secant even, the others odd); Calculator evaluation of trig functions in radian mode, including csc, sec, cot as reciprocals; Estimating sine and cosine values from a unit circle figure, and approximating t for a given value; Evaluating trigonometric models at given times, such as a spring's displacement. **4.3** Cosecant, secant, and cotangent as right-triangle ratios; Reciprocal identities; Quotient identity for cotangent (cot = cos/sin); Pythagorean identities; Cofunction identities beyond the sine-cosine pair (tangent-cotangent, secant-cosecant); Special-angle values in radians (pi/6, pi/4, pi/3), and finding the angle from a value; Finding all six function values from one given ratio; Using fundamental identities to simplify or verify trig expressions; Calculator evaluation of trig functions at non-special acute angles (degree mode, decimal degrees and DMS, reciprocals via 1/x); Finding an acute angle from a known side ratio with a calculator; Solving right triangles at non-special angles (missing sides); Angles of elevation and depression in applied height and distance problems. **4.4** Six trig functions defined from a point (x, y) and r; Cosecant, secant, and cotangent of any angle; Signs of the trig functions by quadrant; Identifying the quadrant from the signs of two functions; Choosing a point on a terminal side given as a line in a stated quadrant; Values at quadrantal angles, including where a function is undefined; Finding the remaining function values from one value and its quadrant; Finding reference angles in degrees and radians; Coterminal step for negative angles and angles past one revolution; Combining a reference-angle value with the quadrant sign; Special-angle values in radians (pi/6, pi/4, pi/3); Finding both angles in one revolution with a given special value; Trig functions of real numbers evaluated with reference angles; Calculator evaluation of trig functions in degree and radian mode, including csc, sec, cot as reciprocals; Evaluating trig models in applied problems. **4.5** Hand sketching a transformed sinusoid: finding the one-cycle interval and the five key points at quarter-period spacing; Rewriting a negative b with the even/odd properties of sine and cosine; Frequency as the reciprocal of the period; Writing a complete sinusoid equation from its graph (choosing sine or cosine form and combining a, b, c, d); Sinusoidal models fitted to real data (finding a, b, c, d from a table or a description). **4.6** Cotangent graph, period, and asymptotes; Secant and cosecant graphs sketched from cosine and sine guide curves; Asymptotes and local maxima and minima of secant and cosecant branches; Vertical stretch of tangent and cotangent graphs, which have no amplitude; Locating two consecutive asymptotes of transformed tangent and cotangent graphs by solving bx - c = -pi/2 and pi/2, or 0 and pi; Phase shifts of tangent, cotangent, secant, and cosecant graphs; Even/odd symmetry seen in the tangent, cotangent, secant, and cosecant graphs; Domain, range, and period summary of all six trigonometric functions; Solving equations such as tan x = c or sec x = c graphically on an interval; Damped trigonometric graphs and damping factors; Bounding curves y = plus or minus f(x) for products such as x sin x; Behavior of functions such as (sin x)/x as x approaches 0. **4.7** Restricting sine, cosine, and tangent to one-to-one intervals; Definitions, domains, and ranges of arcsin, arccos, and arctan; The notation sin^-1 x and why it is not the reciprocal 1/sin x; Exact values of arcsin, arccos, and arctan in radians, including negative and quadrantal inputs; Calculator evaluation of inverse trig functions; Graphs of y = arcsin x, y = arccos x, and y = arctan x, including transformed versions; Inverse properties such as sin(arcsin x) = x and arcsin(sin y) = y, with their interval conditions; Compositions whose input lies outside the restricted domain or range; Exact values of compositions like tan(arccos a) at a number a, from a right-triangle sketch, including negative a; Rewriting compositions such as sin(arctan x) as algebraic expressions in x; Writing a triangle angle as an inverse trig function of x; Rewriting one inverse trig expression as another via a right triangle (exercise extension); Inverse cotangent, secant, and cosecant (exercise extension); Identities such as arcsin(-x) = -arcsin x and arcsin x + arccos x = pi/2; Applied problems that find an angle with an inverse trig function. **4.8** Solving right triangles at non-special angles with a calculator; Finding acute angles with inverse trig functions; Angles of elevation and depression in applied height and distance problems; Two-triangle problems, such as two angles of elevation to the same object; Bearings in navigation (N/S-E/W compass form and three-digit nautical form); Simple harmonic motion models d = a sin(wt) and d = a cos(wt); Frequency of harmonic motion (w / 2pi); Writing a harmonic motion model from displacement at t = 0, amplitude, and period (choosing sine or cosine); Evaluating a harmonic motion model at a given time, including the least positive time when d = 0.

Grade 4

☐	46. Time	prereq	4.1
☐	51. Angle measures	prereq	4.1

Grade 5

☐	13. Add fractions with unlike denominators	prereq	4.1
☐	14. Subtract fractions with unlike denominators	prereq	4.1
☐	21. Multiply fractions	prereq	4.1

Grade 6

☐	10. Unit rate	prereq	4.1
☐	14. Unit conversions in the customary system	prereq	4.1
☐	30. Greatest common factor	prereq	4.1
☐	33. Plot points in all quadrants	prereq	4.1, 4.4

Grade 7

☐	41. Circumference of circles	prereq	4.1, 4.2
☐	42. Area of circles	prereq	4.1
☐	43. Complementary angles	partial	4.1, 4.3, 4.8
☐	44. Supplementary angles	partial	4.1, 4.4

Grade 8

☐	73. Alternate interior angles	prereq	4.8
☐	76. Interior angles of a triangle	prereq	4.8

Grade 9

☐	43. Simplify radicals	prereq	4.3, 4.4

Grade 10

	6. Define rotations	prereq	4.1
	8. 90-degree clockwise rotations	prereq	4.1
	17. Proportional relationships in similar triangles	prereq	4.3
	20. Pythagorean theorem	prereq	4.3, 4.4, 4.7, 4.8
	23. Intro to the sine ratio	full	4.3, 4.7, 4.8
	24. Sine of 30	partial	4.2, 4.3, 4.4
	25. Sine of 60	partial	4.2, 4.3, 4.4
	26. Sine of 45	partial	4.2, 4.3, 4.4
	27. Intro to the cosine ratio	full	4.3, 4.7, 4.8
	28. Cosine of 60	partial	4.2, 4.3, 4.4
	29. Cosine of 30	partial	4.2, 4.3, 4.4
	30. Cosine of 45	partial	4.2, 4.3, 4.4
	31. Intro to the tangent ratio	full	4.3, 4.7, 4.8
	32. Tangent of 45	partial	4.2, 4.3, 4.4
	33. Tangent of 30	partial	4.2, 4.3, 4.4
	34. Tangent of 60	partial	4.2, 4.3, 4.4
	35. Connect sine and cosine	partial	4.3, 4.7
	36. Connect sine, cosine, and tangent	partial	4.2, 4.3, 4.6, 4.7
	41. Reflections across the x-axis as functions	prereq	4.2
	43. Reflections across y=x as functions	prereq	4.7
	47. Circles on the plane	prereq	4.2
	51. Distance formula	prereq	4.4

Grade 11

	42. Exponential decay	prereq	4.6
	70. Horizontal translations	prereq	4.5
	71. X-axis function reflections	partial	4.5, 4.6
	72. Y-axis function reflections	partial	4.5
	77. Intro to the sine function	partial	4.2, 4.5, 4.6, 4.7
	78. Period of sine functions	partial	4.2, 4.5
	79. Intro to the cosine function	partial	4.2, 4.5, 4.6, 4.7
	80. Period of cosine functions	partial	4.2, 4.5
	81. Midline of sine and cosine functions	full	4.5
	82. Amplitude of sine and cosine functions	full	4.5
	83. Intro to the tangent function	partial	4.2, 4.6, 4.7
	84. Period of tangent functions	full	4.6

Grade 12

☐	9. Vertical asymptotes of rational functions	prereq	4.6
☐	13. Function composition	partial	4.7
☐	14. Inverse functions	partial	4.7
☐	17. Domain and range	partial	4.7
☐	18. Graphing inverses	partial	4.7
☐	19. Period of sinusoidal functions	full	4.5, 4.6, 4.8
☐	20. Midline of sinusoidal functions	full	4.5
☐	21. Amplitude of sinusoidal functions	full	4.5, 4.6, 4.8
☐	22. Phase shift of sinusoidal functions	full	4.5, 4.6
☐	23. Relate sine and cosine	partial	4.5, 4.8

Geometry

☐	12. Circles	prereq	4.1
☐	13. Angle relationships	partial	4.1
☐	23. The Pythagorean theorem	prereq	4.3, 4.8
☐	24. Sine ratio	full	4.3, 4.8
☐	25. Cosine ratio	full	4.3, 4.8
☐	26. Tangent ratio	full	4.3, 4.8
☐	27. Sine function	partial	4.2, 4.5
☐	28. Cosine function	partial	4.2, 4.5
☐	29. Tangent function	partial	4.2, 4.6
☐	30. Graph sine and cosine	full	4.5, 4.6, 4.8

Algebraic Graphing

☐	23. Inverse linear functions	prereq	4.7

Textbook: Ron Larson and Robert P. Hostetler. Precalculus with Limits, 2nd ed. (Pre-AP National Student Edition) Thomson Brooks/Cole, 2009. ISBN 978-0-538-73659-6. Duolingo inventory: 820 units, last verified 2026-10-07 on iOS. Only titles and unit names are reproduced, for reference; no textbook or Duolingo content is included. This checklist is licensed CC BY 4.0; see the repository NOTICE for third-party material.